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Robust webly supervised fine-grained recognition via decoupled global-local fusion and geometric-semantic consensus

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ABSTRACT

Webly Supervised Fine-Grained Recognition demands robust learning mechanisms to mitigate the impact of noisy labels and misleading annotations. To address this, current methods typically rely on prediction probabilities to filter noisy labels. However, such paradigms often struggle with mislabeled yet high-confidence samples, leading to confirmation bias. In this paper, we propose a novel framework named Decoupled Global-local Consensus Learning (DGCL) that identifies and leverages noisy-labeled samples through multi-perspective analysis, enhancing both robustness and representation power. Motivated by the high-quality dense representations of foundation models, we first introduce the Decoupled Global-Local Fusion (DGLF) framework, which leverages LoRA to calibrate the encoder's attention, enabling precise identification of discriminative regions. Subsequently, a dual-stream bilinear mechanism is designed to facilitate global-local interactions for enhanced detail capture. To mitigate confirmation bias, we introduce the Geometric-Semantic Consensus (GSC) strategy, which integrates geometric consistency and semantic confidence to partition samples into three mutually exclusive subsets. Finally, a Noise-Aware Supervised Contrastive (NASC) loss is introduced to convert filtered noise into repulsion signals, effectively compressing intra-class variance. Extensive experiments demonstrate that DGCL achieves a new state-of-the-art average accuracy of 91.73% on three web-supervised benchmark datasets, surpassing the previous best method by 1.63 percentage points. Our source code will be made publicly available at: <https://github.com/YT3DVision/DGLF>.

1. Introduction

Fine-Grained Visual Categorization (FGVC) plays a critical role in various real-world applications, ranging from biodiversity monitoring to industrial inspection (Bergmann et al., 2019; Cui et al., 2018; Rodner et al., 2015; Van Horn et al., 2018). However, acquiring large-scale, high-quality manual annotations for such specialized tasks remains prohibitively expensive in terms of both expert labor and annotation cost (Xia et al., 2019). To circumvent this bottleneck, Webly Supervised Learning (WSL) has emerged as a scalable paradigm, utilizing search engines and web-crawling techniques to automate data acquisition. Nevertheless, web datasets inherently suffer from two entangled challenges that impede robust learning: (1) Severe Label Noise, including mislabeled samples as well as out-of-distribution (OOD) instances; and (2) High Inter-Class Similarity, which renders the disambiguation of visually similar hard positives from true noise extremely difficult. This vulnerability is further exacerbated by the “memorization effect” of deep neural networks, which have been shown to memorize even entirely randomized labels (Zhang et al., 2016).

Concurrently, the advent of Visual Foundation Models (VFM), such as DINOv2 (Oquab et al., 2024), has brought a significant shift in visual representation learning. Trained in a self-supervised manner, DINO-based models learn label-agnostic and high-quality dense representations, which have demonstrated strong transferability across a broad range of downstream tasks, including object detection (Li et al., 2023; Liu et al., 2024; Zhang et al., 2022), semantic segmentation (Cheng et al., 2022; Hamilton et al., 2022; Oquab et al., 2024), and dense correspondence (Amir et al., 2021; Caron et al., 2021; Zhang et al., 2023). Compared to traditional supervised backbones that rely on task-specific annotations, DINOv2 captures intrinsic visual primitives and preserves subtle fine-grained structures that are not explicitly encouraged during supervised training. These properties make DINOv2 particularly well suited for web-supervised FGVC, where robustness to label noise and sensitivity to local discriminative details are simultaneously required.

When adapting foundation models to web-supervised FGVC, training feasibility and parameter efficiency become critical bottlenecks. Existing dual-network architectures (Bai et al., 2025; Sheng et al., 2024c) improve robustness to noisy labels through cross-supervision. Given the

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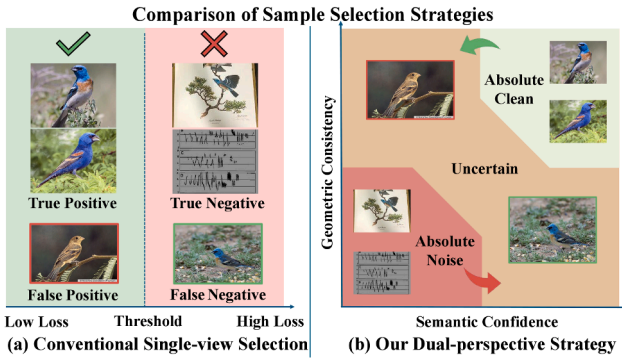


Fig. 1. Motivation of our GSC strategy. (a) Conventional single-view selection relies on the small-loss criterion, leading to confirmation bias: overfitted noise is retained due to low loss (false positives), while hard positives are discarded due to high loss (false negatives). (b) The proposed Geometric-Semantic Consensus (GSC) introduces a geometric view to complement semantic confidence. Geometric inconsistency exposes overfitted noise as uncertain (*Rejected*), while hard positives are rescued into the uncertain set for soft learning due to the strong geometric consistency (*Rescued*).

scale of foundation encoders such as DINOv2 (Oquab et al., 2024), maintaining multiple backbones during training becomes computationally prohibitive. Freezing the backbone leads to a domain gap, while full fine-tuning is both computationally expensive and prone to catastrophic forgetting under noisy supervision. Therefore, we explore a parameter-efficient paradigm using LoRA (Hu et al., 2022) to align these powerful priors with fine-grained task requirements. Nevertheless, mere parameter adaptation is insufficient. Distinguishing fine-grained sub-categories further necessitates capturing high-order interactions between global structures and local discriminative textures. Existing high-order interaction methods (Gao et al., 2016; He et al., 2022; Li et al., 2017; Lin et al., 2015) struggle to scale efficiently to high-dimensional foundation model features due to their reliance on computationally expensive bilinear operations.

Moreover, the narrow decision boundaries inherent in FGVC pose significant challenges for robust representation learning. Despite their progress, most existing selection methods (Han et al., 2018; Li et al., 2020; Sheng et al., 2024b; Wei et al., 2020) remain predicated on the small-loss criterion. This paradigm assumes that samples with lower training loss are more likely to be clean. However, as illustrated in Fig. 1(a), this assumption often fails in fine-grained scenarios characterized by high inter-class similarity. Prior studies on learning with noisy labels have shown that deep networks may eventually memorize mislabeled samples with spurious correlations (Han et al., 2018; Li et al., 2020; Zhang et al., 2016). As a result, some mislabeled samples can exhibit deceptively low losses and high semantic confidence. In contrast, clean hard positives, such as samples with atypical poses or severe occlusions, often remain difficult to optimize and therefore retain relatively higher losses. This mismatch between semantic confidence and sample correctness exposes the limitation of relying solely on loss- or confidence-based sample selection. Similarly, existing contrastive learning strategies (Albert et al., 2022; Wan et al., 2024) attempt to improve robustness to label noise by expanding class boundaries or softening labels; however, such modifications may blur the delicate inter-class margins required for fine-grained discrimination. These limitations necessitate a shift toward a trustworthy identification mechanism grounded in multi-view consensus (Humblot-Renaux et al., 2024), together with a noise-aware optimization objective that repurposes noise as constructive repulsive signals to regularize decision boundaries.

To address these challenges, we propose a novel framework, named Decoupled Global-local Consensus Learning (DGCL), which integrates feature decoupling with consensus-based sample purification. Specifically, first, we propose the Decoupled Global-Local Fusion (DGLF)

framework, which employs LoRA-based attention calibration and a Patch Selection Module (PSM) to extract discriminative regions from foundation model priors with minimal parameter overhead, as shown in Fig. 2. A Dual-Stream Factorized Bilinear Pooling (FBP) module is subsequently introduced to compute the cross-covariance between global and local spaces, explicitly capturing high-order interactions to resolve feature entanglement. Second, to mitigate the confirmation bias illustrated in Fig. 1(a), we propose the Geometric-Semantic Consensus (GSC) strategy. By integrating geometric consistency with semantic confidence, GSC partitions samples into three distinct subsets: *Absolute Clean*, *Uncertain*, and *Absolute Noise*, as depicted in Fig. 1(b). Finally, the Noise-Aware Supervised Contrastive (NASC) loss is formulated to refine decision boundaries, leveraging absolute noise as forced negatives to create a category-agnostic repulsion signal that promotes feature space compactness. Our main contributions are summarized as follows:

- We propose the DGLF framework, which adapts foundation models via LoRA and captures decoupled fine-grained features through a Dual-Stream FBP mechanism, addressing feature ambiguity and inefficiency.
- We introduce the GSC strategy paired with the NASC loss. This multi-view consensus mechanism effectively identifies reliable samples, while the forced negative mining converts absolute noise into beneficial repulsion signals to enhance boundary discriminability.
- Extensive experiments on three web-supervised benchmarks demonstrate that our method outperforms fully fine-tuned and multi-network SOTA methods, achieving the highest average accuracy of 91.73%.

2. Related work

2.1. Noise handling in web-FGVC

Learning fine-grained visual models from web-crawled data inevitably involves handling complex label noise, including closed-set semantic confusion and open-set out-of-distribution (OOD) images. Existing noise handling methods in Web-FGVC can be roughly grouped into four representative directions.

Early approaches primarily focus on filtering out noisy instances based on statistical differentiation. Methods like Jo-SRC (Yao et al., 2021) use Jensen-Shannon (JS) divergence to measure sample likelihood, while PNP (Sun et al., 2022) estimates noise types probabilistically. To address severe class imbalance, SED (Sheng et al., 2024b) introduces dynamic category-specific selection thresholds based on exponential moving averages.

To overcome the confirmation bias inherent in single-network selection, dual-network architectures have been widely adopted. Peer-learning (Sun et al., 2021b) establishes a mutual supervision baseline, whereas recent frameworks like CA2C (Sheng et al., 2025) employ asymmetric co-training with partial label learning and negative learning to mitigate error accumulation.

Instead of simply discarding high-loss samples, another line of research attempts to correct or reuse them. Co-LDL (Sun et al., 2021a) integrates label distribution learning to soften the labels of hard samples. Jo-SNC (Sun et al., 2025) further combines consistency learning with negative learning to extract useful information from noisy samples, while NPPRC (Yuan et al., 2025) introduces noise-free prototypes to guide robust feature alignment.

Recognizing the importance of local details in fine-grained tasks, recent works have shifted the focus towards region-level denoising. HCL (Sun et al., 2023) proposes hierarchical object-part consistency, and DISC (Chen et al., 2025) utilizes bag-level top-down attention to explicitly suppress background noise while emphasizing local discriminative features.

Despite recent progress, two fundamental challenges remain in Web-FGVC. First, although recent methods such as HCL (Sun et al., 2023)

and DISC (Chen et al., 2025) have begun to incorporate region-level cues, image-level reliability estimation remains a dominant component in many existing Web-FGVC frameworks. Such image-level evaluation may inadvertently discard samples that contain informative local details despite cluttered backgrounds, indicating the need for a more decoupled feature representation. Second, although several recent studies attempt to exploit noisy samples through complementary labels or negative learning, OOD instances are still primarily isolated or suppressed. Their potential role beyond noise suppression, particularly as informative signals for feature-space regularization, remains insufficiently explored. These limitations motivate the development of more effective mechanisms for jointly exploiting local discriminative information and OOD supervision.

2.2. Fine-grained visual categorization (FGVC)

FGVC focuses on distinguishing subordinate categories characterized by subtle inter-class differences. To capture these fine-grained details, a significant body of literature on feature representation learning can be broadly grouped into two paradigms: discriminative part localization and high-order feature interaction.

One mainstream direction explicitly locates informative regions to enhance representation. For instance, MC-Loss (Chang et al., 2020) encourages different channels to focus on distinct object regions, while TransFG (He et al., 2022) and ProtoFormer (Xue et al., 2022) integrate Part Selection Modules (PSM) or prototypical learning into Vision Transformers. Recently, MFSL-Net (Yu et al., 2025) utilizes multi-scale spatial attention to adaptively tokenize discriminative regions for noise suppression. While effective on clean benchmarks, these methods typically assume reliable supervision to learn accurate attention. In web-supervised scenarios, inevitable label noise can easily mislead the localization process, causing the model to focus on irrelevant backgrounds rather than the object of interest.

Another stream of research emphasizes modeling feature correlations to capture subtle differences. API-Net (Zhuang et al., 2020) captures attentive pairwise interactions to distinguish confusing classes. Similarly, Bilinear CNN (Lin et al., 2015) and its compact variants, such as CBP (Gao et al., 2016) and FBP (Li et al., 2017), perform second-order pooling to fuse features. However, traditional interaction methods face a dilemma: direct outer products cause dimensional explosion, while simple concatenation leads to semantic ambiguity. To address this issue, recent work, such as VCE (Wang et al., 2025), employs learnable queries to decompose objects into independent semantic units. However, their effectiveness depends on accurate feature alignment, which is vulnerable to noise in webly supervised settings.

3. Method

In web-supervised fine-grained visual categorization (Web-FGVC), given a training set $\mathcal{D} = \{(x_i, \tilde{y}_i) | i = 1, \dots, N\}$, where x_i represents the input image, the corresponding label $\tilde{y}_i \in \{1, \dots, C\}$ always contains significant noise. The core challenge is to learn a robust classifier f_θ capable of distinguishing subtle differences between classes despite the interference of incorrect labels.

As illustrated in Fig. 2, our proposed framework named DGCL includes three components. First, to enhance feature representation capacity (Top Left), we introduce the Decoupled Global-Local Fusion (DGLF) Framework (Sect. 3.1), which leverages LoRA-calibrated attention to identify discriminative regions and constructs the high-dimensional feature representation $z_i = [z_{cls}^i, z_{app}^i, z_{sal}^i]$ via the Patch Selection Module (PSM). z_i is subsequently processed by a dual-stream bilinear mechanism to obtain the predicted results. Second, the Geometric-Semantic Consensus (GSC) Strategy (Sect. 3.2) is proposed for noise classification (Top Right). By aggregating geometric consistency and semantic confidence, GSC dynamically partitions the dataset into *Absolute Clean*, *Uncertain*, and *Absolute Noise* subsets. Finally, a Noise-Aware Learning Mod-

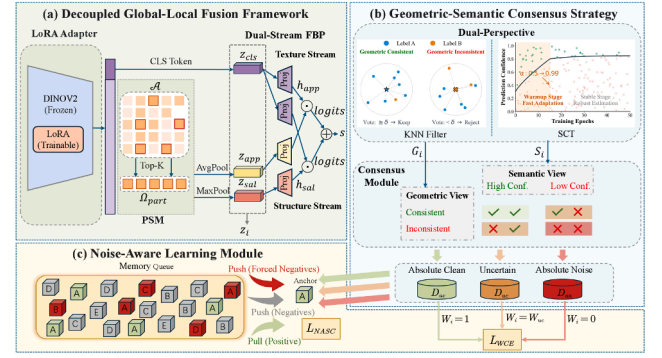


Fig. 2. Overview of the proposed framework DGCL. DGCL consists of three components: (Left) The DGLF module extracts decoupled fine-grained features via LoRA-based attention calibration and dual-stream fusion; (Right) The GSC strategy partitions samples into three reliability subsets based on geometric-semantic consensus; (Bottom) The model is optimized with soft-weighted cross-entropy and the proposed NASC loss, which leverages absolute noise as forced negatives. For ease of visualization, attention vector $\mathcal{A}$ is reshaped into a two-dimensional matrix according to the original spatial arrangement of the image patches.

ule (Bottom) (Sect. 3.3) is proposed to guide the learning of our DGCL. Specifically, to mitigate the impact of label noise, soft re-weighting is applied to uncertain samples, while employing the Noise-Aware Supervised Contrastive (NASC) loss to explicitly leverage noisy labels by converting absolute noise into “forced negative” repulsion signals.

3.1. Decoupled global-local fusion framework

Fine-grained recognition tasks are characterized by high inter-class similarity, where discriminative features are often confined to subtle or highly localized regions. While standard Vision Transformer based networks (backbones), such as DINOv2, possess strong representation capacity, their attention maps tend to be overly dispersed when pre-trained on broad-domain data. As a result, instead of capturing the task-specific traits essential for fine-grained recognition, they often gravitate toward general foreground saliency.

To address this issue, we introduce Low-Rank Adaptation (LoRA) (Hu et al., 2022) to calibrate the attention mechanism while preserving the pre-trained feature space. Formally, we freeze the pre-trained weights $W_0 \in \mathbb{R}^{d \times d}$ of the query and value projection layers in each Transformer block and inject trainable low-rank matrices $A \in \mathbb{R}^{r \times d}$ and $B \in \mathbb{R}^{d \times r}$ (where $r \ll d$). The calibrated weight is formulated as:

$$W' = W_0 + BA. \quad (1)$$

This calibration implicitly encourages the model to suppress background noise and focus more sharply on discriminative fine-grained regions (as visualized in Fig. 3), thus providing more precise feature representations for subsequent processing.

Although the global class token z_{cls} captures robust semantic context, representations from deeper layers of the class token tend to be highly abstract, often attenuating the fine-grained spatial details essential for sub-category disambiguation. To overcome this issue, we propose a Part Selection Module (PSM) that re-excavates diagnostic cues directly from the patch sequence.

To identify discriminative local regions, we extract the attention matrix $Atten \in \mathbb{R}^{N \times N}$ from the final self-attention block of the DINOv2 backbone. We then take the first row of $Atten$, which corresponds to the attention weights from the global class token to all patch tokens. Let $\mathcal{A} \in \mathbb{R}^N$ denote this extracted attention vector. Based on the attention vector $\mathcal{A}$, we use the TOP-K algorithm to get the index array Ω_{part} of K_p patches:

$$\Omega_{part} = \text{TopK}(\mathcal{A}, K_p), \quad (2)$$

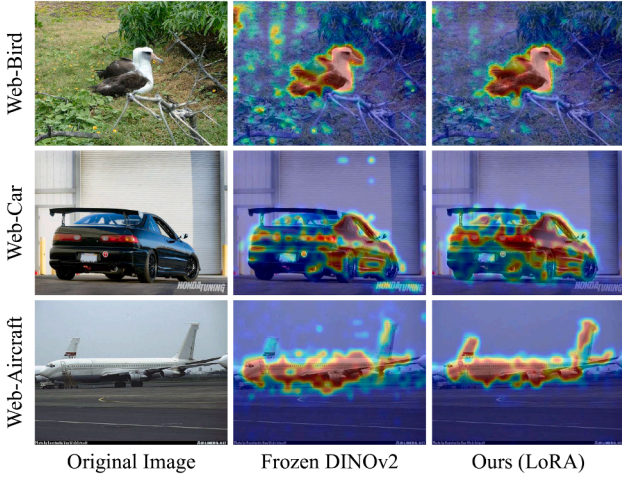


Fig. 3. Effect of attention calibration. Compared with the dispersed attention maps produced by the frozen DINOv2 backbone, which frequently attend to background regions, the proposed LoRA-adapted model suppresses irrelevant distractions and consistently focuses on discriminative object parts (e.g., bird wings and aircraft tails).

thereby suppressing background regions and retaining the most discriminative local features.

To construct a comprehensive local representation, we process the selected parts through two complementary aggregation streams to balance statistical stability and detail sensitivity. While average pooling captures holistic textures (via the Appearance Stream, z_{app}), it often smooths out peak diagnostic signals; conversely, max pooling preserves the most prominent discriminative responses (via the Salient Stream, z_{sal}) but may be susceptible to outliers. By integrating both streams, our strategy provides a more comprehensive characterization of local discriminative evidence. The mathematical formulations for average pooling and max pooling can be expressed as:

$$z_{app} = \frac{1}{K_p} \sum_{j \in \Omega_{part}} v_j, \quad (3)$$

$$z_{sal} = \max_{j \in \Omega_{part}} (v_j), \quad (4)$$

where v_j represents the feature vector of the j -th patch. For the subsequent geometric consistency check in GSC, these components are concatenated to form a unified high-dimensional representation $z_i = [z_{cls}, z_{app}, z_{sal}]$ for the i -th image.

Standard linear fusion primarily captures first-order statistics, which often leads to feature entanglement, where dominant global structures suppress subtle local textures. To address this issue and capture the fine-grained interplay between structure and texture, we propose a dual-stream interaction mechanism based on Factorized Bilinear Pooling (FBP). From a representation learning perspective, FBP computes the cross-covariance matrix between the global and local spaces. This operation explicitly captures high-order feature interactions, mathematically projecting the entangled features into a more separable manifold for fine-grained disambiguation. Taking the Salient Stream as an example, we model the interaction between the global semantics z_{cls} and the salient details z_{sal} to generate the bilinear feature h_{sal} :

$$h_{sal} = \phi((U_{sal} z_{cls}) \odot (V_{sal} z_{sal})), \quad (5)$$

where $U_{sal}, V_{sal} \in \mathbb{R}^{D \times d}$ are learnable projection matrices mapping features to a high-dimensional space $\mathbb{R}^D$, $\odot$ represents the Hadamard product, and $\phi(\cdot)$ denotes the signed square-root followed by ℓ_2 normalization to stabilize the feature distribution. Similarly, we compute the appearance interaction feature h_{app} using the appearance stream z_{app} . The final classification score s is obtained by summing the predictions from

both streams:

$$s = C_{app}(h_{app}) + C_{sal}(h_{sal}), \quad (6)$$

where $C(\cdot)$ denotes the linear classifier for each stream. This decoupled fusion strategy ensures that the model jointly optimizes for both global structure and local texture features.

3.2. Geometric-semantic consensus strategy

Single-view selection methods typically rely on semantic confidence (e.g., loss) to filter samples, a strategy that inherently suffers from confirmation bias, where the model mistakenly reinforces its own erroneous predictions on high-confidence false positives. To mitigate this self-reinforcing error, we propose the Dual-Perspective Geometric-Semantic Consensus (GSC) strategy. From the perspective of multi-view learning, this strategy explicitly decouples the verification process into two orthogonal dimensions: semantic confidence from model predictions and geometric consistency from the feature space. Given the underlying true label, the network's output distribution (the semantic view) and the pre-trained feature manifold (the geometric view) serve as two conditionally independent views. According to the co-training bound, when two such independent views reach a consensus, the joint error rate decreases exponentially compared to a single view. By enforcing this cross-modal agreement, GSC provides a more rigorous screening mechanism, such that samples are selected only when they are supported by both "internal" semantic logic and "external" geometric structure, ensuring that the retained instances are statistically credible rather than merely overfitted noise.

SCT: For the semantic view, we introduce Semantic Confidence Thresholding (SCT). Unlike previous methods (e.g., SCS (Sheng et al., 2024b)), which rely on mean statistics that are sensitive to extremely low-confidence outliers, SCT employs the median to provide a more robust estimation of the global learning state. Furthermore, recognizing the non-linear learning trajectory of models, we introduce a dynamic warmup schedule with linearly increasing momentum. This design enables the threshold to rapidly adapt to the model's convergence in the early stage, while maintaining high stability in the later stage, thereby filtering out batch-wise fluctuations. Let $p_{i, \tilde{y}_i}$ denote the predicted probability associated with the model-predicted label $\tilde{y}_i$. The global threshold $\tau_g^{(t)}$ at epoch t is updated via an Exponential Moving Average (EMA) with a time-varying momentum $\alpha^{(t)}$:

$$\tau_g^{(t)} = \alpha^{(t)} \tau_g^{(t-1)} + (1 - \alpha^{(t)}) \cdot \text{Median}(\{p_{i, \tilde{y}_i}\}_{i=1}^N), \quad (7)$$

where N denotes the number of samples in the current mini-batch. The momentum $\alpha^{(t)}$ linearly increases from an initial value α_0 (e.g., 0.5) to the target momentum m (e.g., 0.999) during the warmup period T_{warmup} :

$$\alpha^{(t)} = \alpha_0 + \frac{t}{T_{warmup}} (m - \alpha_0), \quad \text{for } t \leq T_{warmup}. \quad (8)$$

After the warmup period, we fix $\alpha^{(t)} = m$. Considering the variance in learning difficulty across fine-grained sub-categories, a uniform threshold may unfairly penalize hard classes. To address this class imbalance, we modulate the global threshold via a class-aware factor. Let $\bar{p}_c$ denote the EMA of prediction probabilities for all samples predicted as class c , reflecting the model's confidence in that class. We define the modulation factor σ_c to adaptively modulate the threshold according to class difficulty:

$$\sigma_c = \frac{\bar{p}_c}{\max_{k \in \{1 \dots C\}} (\bar{p}_k)}, \quad (9)$$

$$\tau_c^{(t)} = \tau_g^{(t)} \cdot \sigma_c, \quad (10)$$

where $\tau_c^{(t)}$ represents the specific threshold for class c at epoch t . By construction, $\sigma_c \in (0, 1]$, resulting in a lower threshold for relatively harder classes. Finally, a sample $(x_i, \tilde{y}_i)$ is considered semantically clean ($S_i = 1$)

if its prediction confidence exceeds the corresponding class threshold:

$$S_i = \mathbf{1}_{\{p_{i,\tilde{y}_i} \geq t_{\tilde{y}_i}^{(c)}\}}. \quad (11)$$

KNN Filter: For the geometric view, we introduce a K-Nearest Neighbors (KNN) filter to complement the semantic predictions. While semantic confidence provides valuable category cues, it may suffer from confirmation bias under noisy supervision, particularly in fine-grained scenarios with high inter-class similarity. To provide an independent structural reference, we leverage the high-dimensional features z_i extracted by the DGLF module, which preserve fine-grained visual similarity and facilitate reliable neighborhood modeling. Based on this feature space, a KNN search is performed over the entire training set to characterize the local geometric structure surrounding each sample.

Specifically, for each sample x_i , we retrieve its set of K nearest neighbors $\mathcal{N}_K(i)$ from the entire training set $\mathcal{D}$ based on cosine similarity. It is worth noting that since our framework adopts a two-stage pipeline where features are pre-computed, this global retrieval remains computationally efficient and practical for large-scale datasets. We then calculate the local consistency score c_i by counting the label consistency within the neighborhood:

$$c_i = \sum_{j \in \mathcal{N}_K(i)} \mathbf{1}_{\{\tilde{y}_j = \tilde{y}_i\}}. \quad (12)$$

The geometric filter is based on the manifold assumption, which states that samples from the same semantic class tend to occupy nearby regions in the feature space. Therefore, a sample is considered geometrically reliable when a sufficiently large proportion of its neighbors share the same label. Even when a noisy sample obtains a deceptively high semantic confidence due to semantic prediction bias, it is less likely to satisfy the neighborhood consistency criterion if its local neighborhood contains heterogeneous labels. A sample is considered geometrically clean ($G_i = 1$) if the proportion of consistent neighbors exceeds a predefined ratio δ :

$$G_i = \mathbf{1}_{\left\{\frac{c_i}{K} \geq \delta\right\}}. \quad (13)$$

Dataset Splitting: The two indicators are then jointly utilized to partition the dataset into three mutually exclusive subsets:

(1) *Absolute Clean* $\mathcal{D}_{ac} = \{(x_i, \tilde{y}_i) \mid S_i = 1 \wedge G_i = 1\}$, containing samples unanimously verified by both internal semantic confidence and external geometric structure;

(2) *Absolute Noise* $\mathcal{D}_{an} = \{(x_i, \tilde{y}_i) \mid S_i = 0 \wedge G_i = 0\}$, representing samples unanimously rejected by both views;

(3) *Uncertain* $\mathcal{D}_{uc} = \{(x_i, \tilde{y}_i) \mid S_i \neq G_i\}$, comprising all instances where the two views yield inconsistent predictions. This subset effectively captures ambiguous noise and hard positives, preserving them for subsequent soft-weighted optimization.

This dual-view decision mechanism enables sample reliability to be assessed from complementary semantic and geometric evidence, thereby providing a more refined partition than conventional single-view thresholding strategies.

3.3. Noise-aware learning module

Traditional sample selection methods often suffer from a binary dilemma, where hard but informative samples are either discarded prematurely to ensure purity or retained at the cost of introducing potential noise. To navigate this trade-off, we assign soft weights w_i based on the GSC three-way partitioning results. Specifically, we apply full weight ($w_i = 1.0$) to the absolute clean set $\mathcal{D}_{ac}$, while assigning a reduced weight ($w_i = w_{uc} \in (0, 1)$) to the uncertain set $\mathcal{D}_{uc}$ to attenuate their potential negative impact. Samples in the absolute noise set $\mathcal{D}_{an}$ are discarded ($w_i = 0$). This strategy allows informative hard samples to contribute to training while mitigating noise interference, thereby balancing information preservation and robustness. The classification loss

is defined as:

$$\mathcal{L}_{wce} = -\frac{1}{N} \sum_{i=1}^N w_i \log p_{i,\tilde{y}_i}. \quad (14)$$

Standard Supervised Contrastive Learning (Khosla et al., 2020) does not explicitly account for label quality, treating all samples sharing the same label as positives. This can induce confirmation bias, as noisy anchors may generate erroneous gradients that adversely affect the learned feature space. To address this, we propose the Noise-Aware Supervised Contrastive (NASC) loss, which incorporates “Anchor Purification” and “Negative Utilization”.

First, Anchor Purification restricts the selection of anchor samples to the verified clean set $\mathcal{D}_{ac}$. By anchoring the contrastive optimization on high-fidelity samples, we ensure that the feature alignment is primarily guided by reliable semantic signals.

Second, instead of treating identified noise as useless outliers, we explore its potential as a source of category-agnostic repulsion via Negative Utilization. We maintain a First-In-First-Out queue Q updated by a momentum encoder. Q stores the normalized feature embeddings (keys) of past samples, along with their corresponding labels and subset memberships. Crucially, samples identified as absolute noise ($\mathcal{D}_{an}$) are treated as forced negatives for all anchors. When constructing the positive set $\mathcal{P}(i)$ for an anchor x_i , we strictly mask out any noise-verified samples, regardless of their nominal labels:

$$\mathcal{P}(i) = \{k \in Q \mid \tilde{y}_k = \tilde{y}_i \wedge k \notin \mathcal{D}_{an}\}. \quad (15)$$

To fulfill the Negative Utilization, the explicit negative set $\mathcal{N}(i)$ is constructed by uniting the standard inter-class negatives with all identified absolute-noise instances:

$$\mathcal{N}(i) = \{k \in Q \mid \tilde{y}_k \neq \tilde{y}_i\} \cup \{k \in Q \mid k \in \mathcal{D}_{an}\}. \quad (16)$$

Consequently, the entire queue space can be explicitly decomposed as $Q = \mathcal{P}(i) \cup \mathcal{N}(i)$.

Following the standard InfoNCE formulation, the probability of matching an anchor q_i with a positive sample k_+ is defined over the entire queue space Q . By explicitly incorporating absolute-noise samples into the negative set $\mathcal{N}(i)$, the resulting Noise-Aware Supervised Contrastive (NASC) objective is derived as:

$$\mathcal{L}_{nasc} = \sum_{i \in \mathcal{D}_{ac}} \frac{-1}{|\mathcal{P}(i)|} \sum_{k_+ \in \mathcal{P}(i)} \log \frac{\exp(q_i \cdot k_+ / \tau)}{\sum_{k \in Q} \exp(q_i \cdot k / \tau)}, \quad (17)$$

where q_i denotes the normalized feature representation of the current anchor sample, k_+ represents the positive key embeddings in $\mathcal{P}(i)$, and τ is a temperature hyper-parameter scaling the distribution sharpness.

By incorporating the identified absolute-noise instances into the contrastive objective, NASC introduces additional repulsive gradients during optimization, discouraging anchors from approaching ambiguous noise regions and helping maintain discriminative decision boundaries.

Finally, the total objective is formulated as $\mathcal{L}_{total} = \mathcal{L}_{wce} + \lambda \mathcal{L}_{nasc}$, where λ balances the classification and contrastive terms. By jointly minimizing $\mathcal{L}_{total}$, the framework updates the trainable parameters, specifically the Dual-Stream FBP classification head and the contrastive projection head. This optimization completes the symbiotic loop between representation learning and sample purification: while the decoupled features from DGLF guide the GSC sample partitioning, the purified supervision in turn protects the FBP module from overfitting to noisy labels, enabling it to generate increasingly precise semantic probabilities for subsequent consensus.

4. Experiments

4.1. Experimental setup

Datasets: We evaluate our method on three widely used webly supervised fine-grained benchmarks: Web-Bird, Web-Car, and Web-Aircraft (Sun et al., 2021b). Following the standard protocol, the noisy

Table 1

Performance comparison with state-of-the-art methods on three webly supervised fine-grained benchmarks. Top-1 classification accuracy (%) is reported on the clean test sets. Methods are grouped into prior approaches and our framework with different backbones. The best results are highlighted in **bold**, and the second-best values are underlined.

Method	Publication	Web-Bird	Web-Car	Web-Aircraft	Avg.
Co-teaching (Han et al., 2018)	NeurIPS 2018	76.70	85.00	79.50	80.40
Co-teaching + (Yu et al., 2019)	ICML 2019	70.10	76.80	74.80	73.90
Jo-CoR (Wei et al., 2020)	CVPR 2020	79.20	85.10	80.10	81.50
Jo-SRC (Yao et al., 2021)	CVPR 2021	81.20	88.10	82.70	84.00
Co-LDL (Sun et al., 2021a)	TMM 2021	81.00	89.20	83.80	84.70
PNP (Sun et al., 2022)	CVPR 2022	81.90	90.10	85.50	85.80
MS-DeJOR (Cai et al., 2023)	PR 2023	83.70	88.40	88.50	86.90
HCL (Sun et al., 2023)	TMM 2023	86.10	91.60	92.50	90.10
CECL (Wan et al., 2024)	AAAI 2024	83.87	87.46	87.46	86.36
NPN (Sheng et al., 2024a)	AAAI 2024	79.36	85.46	83.65	82.82
SED (Sheng et al., 2024b)	ECCV 2024	82.00	88.88	86.62	85.83
ACT (Sheng et al., 2024c)	ACM MM 2024	81.43	88.75	86.56	85.58
DSpace (Pan et al., 2025)	AAAI 2025	80.95	89.45	89.92	86.77
CA2C (Sheng et al., 2025)	ICCV 2025	82.48	89.11	87.70	86.43
Jo-SNC (Sun et al., 2025)	TPAMI 2025	82.76	90.56	87.97	87.10
CDgAM (Zhou et al., 2026)	AAAI 2026	82.73	90.89	86.77	86.80
DINOv2 (Oquab et al., 2024)	TMLR 2024	85.35	81.93	73.90	80.39
DINOv3 (Siméoni et al., 2026)	TMLR 2026	86.95	91.03	81.34	86.44
Ours (DINOv2)	–	89.42	<u>92.26</u>	92.38	<u>91.35</u>
Ours (DINOv3)	–	<u>89.09</u>	92.92	93.19	91.73

web-crawled training sets and clean test sets from the corresponding benchmarks (CUB-200-2011, Stanford Cars, and FGVC-Aircraft) are directly adopted. Web-Bird contains 18,388 training images from 200 categories, Web-Car contains 21,448 images from 196 categories, and Web-Aircraft contains 13,503 images from 100 categories. According to the dataset creators' random sampling estimation (Sun et al., 2021b), the approximate training-set purities are 65%, 67%, and 73% for Web-Bird, Web-Car, and Web-Aircraft, respectively. These webly collected datasets contain both closed-set label noise and open-set out-of-distribution (OOD) samples, making them particularly challenging benchmarks for noise-robust FGVC.

To further validate robustness under controlled noise settings, we additionally conduct experiments on CIFAR-100N and CIFAR-80N (Wei et al., 2022). CIFAR-100N is evaluated under both symmetric (Sym) and asymmetric (Asym) label noise, where the latter introduces semantically similar label corruption. CIFAR-80N further includes an open-set setting in which 20% of the training samples originate from out-of-distribution (OOD) classes outside the target 80 categories.

Implementation Details: Our framework is implemented in PyTorch and trained on a single NVIDIA RTX 4090 GPU. The training process consists of two stages:

1) *Backbone Fine-tuning:* We adopt DINOv2 ViT-L/14 as the visual backbone and initialize it with the official pre-trained weights released by Meta, which were trained on the LVD-142M dataset. We apply LoRA ($r = 16$, $\alpha = 16$, dropout rate = 0.1) to the attention layers. The rank $r = 16$ is selected based on the sensitivity analysis provided in Appendix F. The model is fine-tuned for 15 epochs using AdamW with a backbone learning rate of 1×10^{-4} , a head learning rate of 1×10^{-3} and a weight decay of 1×10^{-4} . The batch size is set to 8. Input images are resized to a resolution of 518×518 and undergo standard data augmentation, including random horizontal flipping and AutoAugment (ImageNet policy). A standard Cross-Entropy loss is used during this stage.

2) *Feature Fusion and Classifier Training:* With the backbone frozen, we extract 3072-dimensional composite features (global class token, average-pooled appearance, and max-pooled salient features) via PSM. Since the Top-K operation is applied only after the feature extraction stage and the corresponding feature extractor remains frozen, no gradient propagation through the Top-K indices is required. We empirically set the number of selected patches to $K_p = 200$ throughout all experi-

ments. The Dual-Stream FBP head ($D = 4096$, dropout = 0.5) and classifier are trained for 50 epochs using the AdamW (learning rate 1×10^{-3} , weight decay 1×10^{-5}). The batch size is set to 128. For GSC, the KNN neighborhood size is set to $K = 10$ with $\delta = 0.5$. The SCT momentum linearly increases from 0.5 to 0.999 during a 10-epoch warm-up period to stabilize EMA threshold estimation during the early stage of training. For NASC, the queue size is set to $N_q = 4,096$. The objective function combines weighted Cross-Entropy ($\omega_{uc} = 0.5$) and NASC ($\lambda = 0.5$) losses. Detailed sensitivity analyses of the LoRA rank and other key hyperparameters (including K , δ , ω_{uc} and λ) are provided in Appendix F.

Baselines. Additionally, to demonstrate the effectiveness of our adaptation framework, we establish a Baseline (DINOv2) (Oquab et al., 2024), which refers to training a simple linear classifier on the frozen DINOv2 features using standard Cross-Entropy loss without any noise purification strategies. To evaluate the scalability of our method to stronger visual foundation models, we additionally report results using DINOv3 under the same protocol.

4.2. Performance comparison

Results on Webly Supervised FGVC. Table 1 reports the performance comparison on the Web-Bird, Web-Car, and Web-Aircraft benchmarks. Unless otherwise specified, all comparisons are conducted using DINOv2 for fair evaluation. The proposed framework achieves the best overall performance with an average accuracy of 91.35%, surpassing the previous state-of-the-art method HCL (Sun et al., 2023) by 1.25%. Compared with the frozen DINOv2 baseline, our framework improves the accuracy by 4.07%, 10.33%, and 18.48% on Web-Bird, Web-Car, and Web-Aircraft, respectively, demonstrating its effectiveness in handling real-world web noise. Furthermore, replacing DINOv2 with DINOv3 further improves the average accuracy to 91.73%, indicating good generalization to stronger visual foundation models.

Results on Controlled Noisy Benchmarks. Table 2 summarizes the results on CIFAR-100N and CIFAR-80N under various synthetic and open-set noise settings. The proposed framework consistently outperforms recent noise-robust learning methods across all evaluated scenarios. Notably, under the challenging Asym-40% setting, our method achieves gains of 21.04% and 22.09% over the frozen DINOv2 baseline on CIFAR-100N and CIFAR-80N, respectively. Furthermore, competitive performance is maintained even under severe corruption levels

Table 2

Performance comparison on synthetic noisy datasets (CIFAR-100N and CIFAR-80N). Top-1 classification accuracy (%) is reported. The best results are highlighted in **bold**, and the second-best values are underlined.

Method	Publication	CIFAR-100N			CIFAR-80N		
		Sym-20%	Sym-80%	Asym-40%	Sym-20%	Sym-80%	Asym-40%
Co-teaching (Han et al., 2018)	NeurIPS 2018	43.73	15.15	28.35	60.38	16.59	42.42
Co-teaching + (Yu et al., 2019)	ICML 2019	49.27	13.44	33.62	53.97	12.29	43.01
JoCoR (Wei et al., 2020)	CVPR 2020	53.01	15.49	32.70	59.99	12.85	39.37
Jo-SRC (Yao et al., 2021)	CVPR 2021	58.15	23.80	38.52	65.83	29.76	53.03
PNP (Sun et al., 2022)	CVPR 2022	64.25	31.32	60.25	67.00	34.36	61.23
CECL (Wan et al., 2024)	AAAI 2024	69.20	36.37	65.49	77.23	37.21	68.48
SED (Sheng et al., 2024b)	ECCV 2024	66.50	38.15	58.29	69.10	42.57	60.87
DSpace (Pan et al., 2025)	AAAI 2025	67.11	48.33	65.22	69.61	49.30	67.27
CA2C (Sheng et al., 2025)	ICCV 2025	68.64	40.97	65.59	70.06	40.47	65.71
Jo-SNC (Sun et al., 2025)	TPAMI 2026	65.49	43.45	61.59	67.81	41.10	62.57
DINOV2 (Oquab et al., 2024)	TMLR 2024	90.74	77.81	66.04	91.42	80.14	67.61
DINOV3 (Siméoni et al., 2026)	TMLR 2026	91.13	88.61	77.50	92.67	88.92	78.04
Ours (DINOV2)	–	95.50	<u>91.73</u>	87.08	95.74	<u>92.86</u>	89.70
Ours (DINOV3)	–	<u>94.82</u>	93.44	<u>85.74</u>	<u>95.64</u>	92.91	<u>89.69</u>

Table 3

Quantitative evaluation of noise filtering precision.

Dataset	Noise Setting	Recall (%)	Single-view Prec. (%)	Consensus Prec. (%)	Consensus F1 (%)
CIFAR-100N	Sym-0.2	99.94	76.04	92.99	96.34
	Sym-0.8	99.63	89.69	99.23	99.43
	Asym-0.4	68.43	67.43	89.09	77.41
CIFAR-80N	Sym-0.2	99.89	89.69	97.79	98.83
	Sym-0.8	99.42	92.32	99.63	99.52
	Asym-0.4	90.43	69.28	95.56	92.93

(e.g., Sym-80%), indicating stable performance across diverse noise conditions.

4.3. Noise filtering robustness

We quantitatively evaluate the noise filtering precision of the GSC strategy. Table 3 compares the filtering precision using a single-view decision boundary (semantic view alone) against our dual-perspective consensus approach. As shown in Table 3, while the single-view approach achieves acceptable precision under symmetric noise, its performance degrades substantially under asymmetric noise. In particular, under Asym-0.4, the consensus mechanism improves filtering precision from 67.43% to 89.09% on CIFAR-100N and from 69.28% to 95.56% on CIFAR-80N. Although the recall decreases on CIFAR-100N under asymmetric noise, this behavior is expected because the consensus strategy intentionally prioritizes precision over coverage. Samples that fail consensus verification are retained in the Uncertain subset and subsequently handled through the soft-weighting mechanism.

4.4. Open-set noise generalization

To evaluate the capability of the proposed framework in handling open-set noise, we analyze its OOD detection performance on the CIFAR-80N dataset. As shown in Table 4, the proposed framework demonstrates strong and stable open-set discrimination capability under different noise settings. Under the Asym-0.4 setting, where visually similar categories make the distinction between hard positives and noisy instances particularly difficult, the AUROC remains at 87.95%. Although the ID detection accuracy decreases to 84.50% due to the increased ambiguity among in-distribution samples, the OOD detection accuracy remains exceptionally high at 99.89%. These results indicate that the proposed geometric-semantic consensus effectively separates OOD instances from the training distribution, while maintaining robust performance across diverse noise configurations.

We further compare DGCL with representative noise-robust learning methods under the Sym-80% setting. As shown in Table 5, DGCL sub-

Table 4

Open-set robustness on CIFAR-80N. We report the OOD detection capability (AUROC) and the corresponding ID and OOD detection accuracy.

Noise Setting	AUROC	ID Det. Acc.	OOD Det. Acc.
Sym-0.2	95.88%	99.95%	99.84%
Sym-0.8	89.71%	99.48%	99.22%
Asym-0.4	87.95%	84.50%	99.89%

Table 5

OOD detection comparison on CIFAR-80N under the Sym-80% setting. We report AUROC and the corresponding ID and OOD detection accuracy.

Method	AUROC	ID Det. Acc.	OOD Det. Acc.
SED	62.68%	89.30%	86.86%
DSpace	60.14%	94.89%	92.73%
Ours	89.71%	99.48%	99.22%

stantially outperforms SED and DSpace in OOD detection, improving AUROC from 62.68% and 60.14% to 89.71%, respectively. Meanwhile, DGCL also achieves higher ID and OOD detection accuracy, demonstrating stronger robustness to open-set noise.

4.5. Ablation studies and analysis

Component-wise Ablation. We first evaluate the individual contributions of the proposed modules across the three WebFG benchmarks (Table 6). Starting from a frozen DINOV2 backbone, each component consistently improves the overall performance, and the complete framework achieves the best average accuracy of 91.35%.

Foundation Model Adaptation: The *Baseline (Frozen)* achieves a relatively low average accuracy of 80.39%. Introducing LoRA improves the average accuracy by 6.39%, suggesting that parameter-efficient

Table 6

Ablation studies on three webly supervised benchmarks using DINOv2 as the backbone. “Avg.” denotes the average accuracy across three datasets, and “w/o” indicates removing the corresponding module or reverting to a baseline setting.

Method Variants	Web-Bird	Web-Car	Web-Air	Avg.
Baseline	85.35	81.93	73.90	80.39
Baseline (LoRA)	87.47	86.38	86.50	86.78
w/o LoRA	88.90	86.95	79.72	85.19
w/o PSM	89.06	91.29	88.72	89.69
w/o FBP	87.49	90.39	90.46	89.45
w/o DGLF	87.37	90.81	87.82	88.67
DGLF $\rightarrow$ MLP	88.19	90.24	91.27	89.90
w/o KNN	89.06	91.73	91.96	90.92
w/o SCT	88.11	91.90	91.93	90.65
w/o GSC	87.52	91.56	91.81	90.30
w/o Median Anchor	89.14	91.63	92.26	91.01
w/o W_{uc}	88.36	91.94	91.78	90.69
w/o W_{an}	88.57	91.71	91.93	90.74
w/o NASC	89.04	91.49	92.26	90.93
w/o OOD Handling	89.01	91.93	92.20	91.05
Ours	89.42	92.26	92.38	91.35

adaptation helps mitigate the domain gap between generic pre-training and noisy fine-grained tasks.

DGLF Framework: Removing the entire DGLF framework reduces the average accuracy to 88.67%. Within DGLF, replacing the Dual-Stream FBP with simple concatenation (w/o FBP) decreases the average accuracy from 91.35% to 89.45%, and replacing the FBP module with an MLP (DGLF $\rightarrow$ MLP) of similar parameter capacity decreases the average accuracy by 1.45%, indicating that effective global-local feature fusion constitutes an important component of the DGLF framework.

GSC Strategy: Disabling the noise purification strategy (w/o GSC) reduces the average accuracy to 90.30%. Using only semantic confidence (w/o KNN) and only geometric consistency (w/o SCT) achieves average accuracies of 90.83% and 90.90%, respectively, both lower than the full model (91.35%). This suggests that the geometric and semantic views provide complementary information for noise estimation. Moreover, the proposed median-based strategy consistently outperforms the mean-based variant (w/o Median Anchor) across all three fine-grained datasets, improving the average accuracy from 91.01% to 91.35%.

Optimization Objectives: Replacing soft re-weighting with hard selection (w/o W_{uc}) or removing the contrastive loss (w/o NASC) decreases the average performance by 0.66% and 0.42%, respectively. Assigning a small non-zero weight (set to 0.1) to samples identified as Absolute Noise (w/o W_{an}) or removing the out-of-distribution treatment in the NASC loss (w/o OOD Handling) consistently degrades the overall performance. These results suggest that retaining uncertain samples through soft weighting and introducing contrastive regularization both contribute to the final performance.

Noise Robustness Analysis of GSC. To analyze the behavior of the GSC module under varying noise intensities, we conduct additional evaluations on the CIFAR-80N dataset with asymmetric noise intensities of 10%, 20%, and 30% (Table 7). As the asymmetric noise intensity increases from 10% to 30%, the performance gap between the full model and the GSC-related variants gradually enlarges. For example, completely removing GSC causes an accuracy drop of 0.69% under 10% noise, which extends to a 4.94% drop under 30% noise. This trend suggests that the contribution of the GSC module becomes increasingly important under more severe label corruption.

4.6. Parameter sensitivity analysis

We further analyze the sensitivity of three key hyperparameters under a unified parameter space: the clean-rate threshold δ , the uncertain-

Table 7

Analysis of GSC under varying asymmetric noise intensities on CIFAR-80N.

Method	Asym-10%	Asym-20%	Asym-30%
w/o KNN	95.74%	93.61%	92.35%
w/o SCT	95.66%	93.10%	92.09%
w/o GSC	95.10%	91.91%	87.80%
Ours	95.79%	93.80%	92.74%

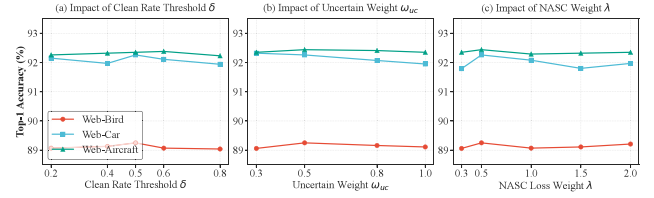


Fig. 4. Parameter sensitivity analysis across different datasets under a unified parameter space. The framework exhibits highly stable performance trends without sudden collapse.

sample weight ω_{uc} , and the NASC loss weight λ . As shown in Fig. 4, DGCL exhibits highly stable performance across different datasets under the same default settings. Although dataset-specific tuning can provide marginal performance gains, the performance gap between the dataset-specific optimum and the unified setting remains below 0.15% on Web-Car and 0.5% on Web-Aircraft. These results demonstrate that DGCL is insensitive to moderate parameter variations and generalizes well without extensive hyperparameter tuning. Based on these observations, we adopt $\delta = 0.5$, $\omega_{uc} = 0.5$, and $\lambda = 0.5$ as the default configuration throughout the experiments. Detailed sensitivity analysis of the geometric parameters K and δ is provided in Appendix F.

4.7. Efficiency analysis

Table 8 compares the computational complexity and inference efficiency of the proposed framework with representative baselines. Although the proposed framework is built upon the DINOv2 foundation model and therefore has a larger total parameter count, only 21.56 M parameters remain trainable due to the frozen backbone design. As shown in Table 8, the trainable parameter count is substantially lower than those of representative dual-network and tri-network methods. A detailed module-wise parameter breakdown is provided in Appendix A.

In terms of deployment efficiency, the proposed framework introduces only 0.02 G additional FLOPs compared with the frozen DINOv2 baseline. Since the KNN filter and NASC queue are used only during training, the inference latency remains nearly unchanged (39.46 ms vs. 39.33 ms). These results suggest that the proposed framework maintains competitive deployment efficiency while incorporating additional noise-robust learning components.

4.8. Cross-backbone generalization

To assess whether the proposed framework generalizes beyond Vision Transformers, we further evaluate it on a ResNet-50 backbone. Since ResNet-50 does not provide discrete patch tokens, we replace the patch-based feature extraction component with global average pooling and spatial max pooling features extracted from the final convolutional stage, while preserving the remaining framework design. For consistency, the baseline ResNet-50 follows the same data augmentation and optimization protocol whenever applicable.

As shown in Table 9, the proposed framework improves the ResNet-50 baseline across all three benchmarks, achieving absolute gains of 12.34%, 10.98%, and 9.09% on Web-Bird, Web-Car, and Web-Aircraft,

Table 8
Comparison of computational complexity and inference efficiency.

Method	Backbone	Total Params	Trainable Params	FLOPs	Inference Latency
Baseline	ResNet-50 $\times$ 1	25.56 M	25.56 M	16.44 G	5.67 ms
TSSD, ACT	ResNet-50 $\times$ 2	51.11 M	51.11 M	32.88 G	11.65 ms
HCL	ResNet-50 $\times$ 3	76.67 M	76.67 M	49.32 G	17.51 ms
Baseline	DINOv2	304.57 M	0.205 M	362.05 G	39.33 ms
Ours (DGCL)	DINOv2 + Modules	325.93 M	21.56 M	362.07 G	39.46 ms

Table 9
Cross-backbone generalization on ResNet-50. Top-1 accuracy (%) on three WebFG benchmarks.

Method	Web-Bird	Web-Car	Web-Aircraft	Average
ResNet50	65.14	75.95	78.22	73.10
Ours (ResNet50)	77.48	86.93	87.31	83.91
Improvement	+12.34	+10.98	+9.09	+10.81

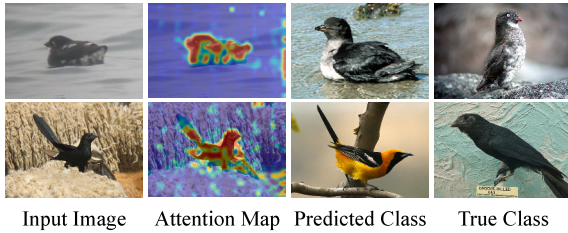


Fig. 5. Failure cases.

respectively. These results indicate that the proposed framework is not restricted to transformer-based architectures and can be effectively transferred to a representative CNN backbone.

4.9. Limitations and failure cases

While effective, our method does have limitations. As shown in Fig. 5, DGCL may still fail under particularly challenging visual conditions. Specifically, the first scene illustrates a failure case where severe blur hinders the identification of discriminative fine-grained details, while the visually similar predicted and true classes further exacerbate the ambiguity. The second scene illustrates a case with strong background clutter and low foreground-background contrast, where discriminative object cues may be obscured by complex textures. In such situations, unreliable attention localization and degraded neighborhood consistency estimation can lead to incorrect geometric-semantic consensus decisions. These observations highlight that improving robustness under severe image degradation and complex backgrounds remains an important direction for future research.

5. Conclusion

In this paper, we propose the Decoupled Global-Local Consensus Learning (DGCL) framework for fine-grained visual categorization, aiming to tackle the dual challenges of high inter-class similarity and complex web noise that contains both in-distribution and out-of-distribution instances. Specifically, DGCL captures decoupled global-local interactions via parameter-efficient adaptation, eliminates confirmation bias through a multi-view geometric-semantic consensus, and repurposes absolute noise into repulsive signals. Extensive experiments on multiple public datasets demonstrate the superiority and practicality of the proposed framework.

CRedit authorship contribution statement

Xinyi Guo: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Conceptualization; **Yiwei Lu:** Writing – review & editing; **Tao Yan:** Writing – review & editing, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The WebFG-469 dataset is publicly available at <https://github.com/NUST-Machine-Intelligence-Laboratory/weblyFG-dataset>.

Appendix A. Model parameter analysis

To provide a detailed parameter breakdown, Table A.1 reports the total and trainable parameters of each module. By freezing the DINOv2 backbone and adopting lightweight adaptation modules, only 21.56 M parameters are involved in optimization, which is substantially smaller than the total model size.

Table A.1

Detailed parameter breakdown of our proposed framework. The pre-trained weights of the DINOv2 backbone are frozen during training.

Component	Total Params (M)	Trainable Params (M)
DINOv2 Backbone [†]	304.37	0
LoRA ($r = 16$)	3.15	3.15
FBP (Appearance)	9.21	9.21
FBP (Saliency)	9.21	9.21
NASC Head	2.36	2.36
Total	325.93	21.56

Appendix B. Geometric feature space analysis

In the main text, geometric consistency in GSC is estimated based on cosine similarity computed from the high-dimensional features z_i . To examine the contribution of each component, we compare the geometric consistency performance obtained using individual feature representations and the fused representation across all three WebFG datasets.

As shown in Table B.1, no single feature component consistently outperforms the others. While individual feature components achieve competitive performance, the concatenated representation consistently achieves the highest accuracy on all three datasets. This supports the use of the combined feature space for geometric consistency estimation in GSC.

Table B.1

Feature representation analysis for geometric consistency estimation. We compare the Top-1 accuracy (%) across three webly supervised fine-grained benchmarks when different feature representations are used to construct the KNN-based geometric space.

Feature Components	Web-Bird	Web-Car	Web-Aircraft	Avg.
Only z_{cls}	88.87	92.03	92.17	91.02
Only z_{app}	88.38	91.95	92.26	90.86
Only z_{sal}	88.70	91.48	92.17	90.78
Ours (z_i)	89.42	92.26	92.38	91.35

Appendix C. Dynamics of GSC sample partitioning

Fig. C.1 presents the evolution of sample allocation among the clean, uncertain, and noise subsets during training. The subset proportions undergo substantial redistribution during the first 15 epochs, indicating that the partitioning criterion rapidly adjusts sample categorization in the early stage of training.

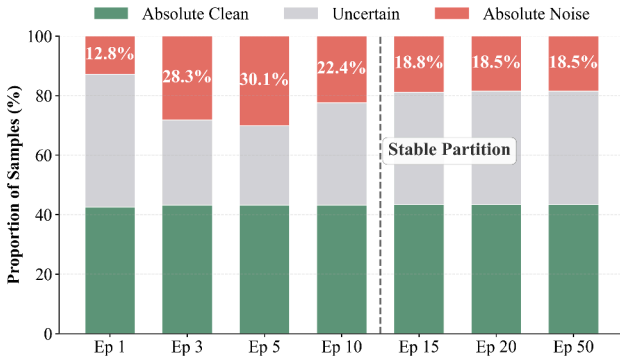


Fig. C.1. Dynamics of GSC sample partitioning across training epochs. The subset proportions rapidly converge within the first 15 epochs and remain stable thereafter.

Afterward, the partition gradually converges and remains stable for the remainder of the optimization process, with no noticeable oscillation in subset proportions. This observation indicates that the resulting sample allocation becomes consistent after the initial adjustment phase.

Interpretability of GSC Partitioning: Furthermore, we provide qualitative examples in Fig. C.2 to illustrate how the Geometric-Semantic Consensus (GSC) strategy robustly partitions samples by leveraging DGLF attention.

For the *Overfitted Noise* instance (Fig. C.2a), the semantic classifier (SCT) produces a high confidence score for the incorrectly given label (Laysan Albatross), suffering from confirmation bias. However, our DGLF module focuses on the discriminative body structure, enabling the KNN geometric view to retrieve visually and semantically identical neighbors (Black-footed Albatross). Since the labels of these neighbors mismatch the given noisy label, the geometric filter correctly exposes this inconsistency, assigning the sample to the Uncertain subset and preventing the model from memorizing the wrong label.

Conversely, Fig. C.2b presents a *Hard Positive* sample characterized by a cluttered background containing multiple birds in rare poses. Consequently, the SCT yields a low confidence score, erroneously treating it as noise. Despite the complex scene, the DGLF attention map captures the key discriminative regions around the bird's body and wing structures. Guided by these informative local features, the KNN retrieves neighbors that share consistent geometric structures and match the given label. This geometric consistency successfully rescues the hard positive sample from being discarded, preserving it in the Uncertain subset for soft-weighted optimization.

Finally, for the OOD Noise (Fig. C.2c), the DGLF attention becomes spatially diffuse and fails to concentrate on semantically meaningful object regions, leading to both a semantic rejection (low confidence) and a chaotic, label-scattered geometric neighborhood. This unanimous rejection assigns the instance to the Absolute Noise subset, safely isolating it to serve as a forced negative. For visualization purposes, we intentionally select a representative OOD example with a clear semantic mismatch map. Overall, these qualitative observations provide intuitive evidence that the proposed dual-consensus mechanism effectively complements semantic confidence with geometric consistency, thereby facilitating more reliable sample partitioning under noisy supervision.

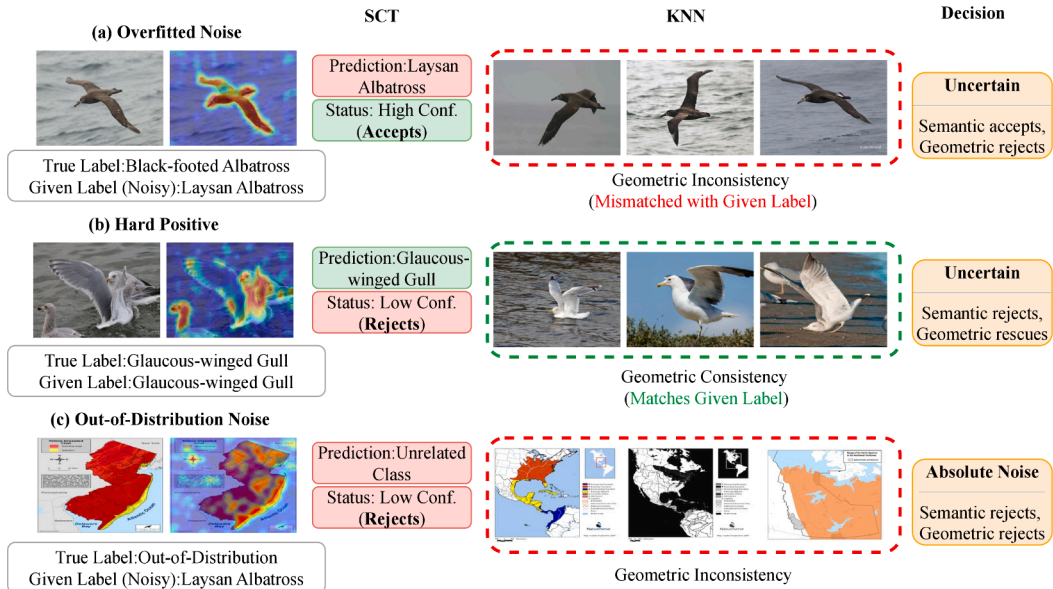


Fig. C.2. Qualitative examples of the GSC partitioning mechanism. (a) An overfitted ID noise sample is exposed by geometric inconsistency. (b) A hard positive sample, initially rejected by SCT due to a cluttered background and a rare pose, is rescued by strong geometric consistency driven by attention on key discriminative regions. (c) An OOD sample is unanimously rejected and assigned to Absolute Noise.

Appendix D. Feature space visualization

t-SNE Visualizations. To evaluate whether the NASC loss effectively regularizes the feature space, we visualize the learned representations using t-SNE. As shown in Fig. D.1, we present the feature distributions on the Web-Bird and Web-Car benchmarks, which are representative real-world Web-FGVC datasets. Compared with the frozen DINOv2 baseline, DGCL without NASC already improves inter-class separability; however, several fine-grained categories still exhibit noticeable overlap. After incorporating NASC, the resulting feature space becomes substantially more compact within each class while maintaining clearer margins between visually similar categories. These observations suggest that the additional repulsive supervision aids in improving cluster compactness beyond the representation provided by DGLF alone. A similar trend can also be observed in the t-SNE visualizations of the dataset CIFAR-80N under Asym-0.4 noise setting (Fig. D.2), further confirming that NASC consistently promotes tighter intra-class clustering and enhanced inter-class discrimination across both synthetic noisy benchmarks and real-world Web-FGVC scenarios.

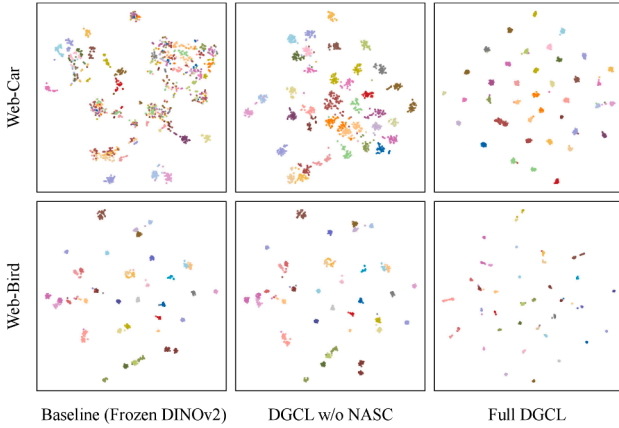


Fig. D.1. t-SNE visualizations of the feature space on Web-Bird and Web-Car. From left to right: Baseline (Frozen DINOv2), DGCL w/o NASC, and full DGCL. Compared with the baseline and DGCL w/o NASC, the full DGCL framework yields tighter intra-class clustering and clearer inter-class separation across real-world Web-FGVC scenarios.

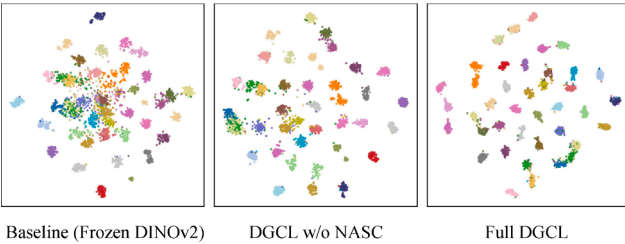


Fig. D.2. t-SNE visualizations of the feature space on CIFAR-80N (40 sampled classes under Asym-0.4 noise). (a) Baseline (Frozen DINOv2) shows severe feature entanglement. (b) DGCL w/o NASC improves separability but exhibits partial overlap. (c) Full DGCL yields more compact intra-class clusters and distinct inter-class margins.

Visualization of Attention Maps at Different Layers. As shown in Fig. D.3, attention in shallow layers (e.g., Layers 6 and 12) remains relatively scattered, capturing both foreground and background contexts. However, as the network deepens (e.g., Layers 18 and 24), the injected LoRA modules effectively steer the feature extraction, progressively suppressing background noise and strictly calibrating the focus towards task-specific discriminative regions.

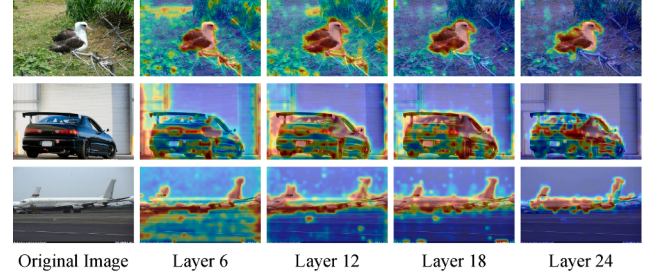


Fig. D.3. Visualization of attention calibration across different transformer layers. From left to right: Original Image, Layer 6, Layer 12, Layer 18, and Layer 24. From top to bottom: Examples from Web-Bird, Web-Car, and Web-Aircraft datasets. As the network deepens, the attention effectively shifts from scattered background context to highly discriminative foreground regions.

Appendix E. Cross-dataset transfer evaluation

To further evaluate robustness under distribution shifts, we conduct an additional cross-dataset transfer experiment from Web-Bird to NABirds (Van Horn et al., 2015). The model is trained solely on Web-Bird with 200 source classes and directly evaluated on NABirds, which contains 555 fine-grained bird categories and 24,633 samples, without any target-domain fine-tuning. Since the Web-Bird and NABirds label spaces are not perfectly aligned, each NABirds category is mapped to its most frequently predicted Web-Bird category based on validation predictions, following a majority-voting strategy for label-space alignment.

Table E.1

Cross-dataset transfer evaluation from Web-Bird to NABirds without target-domain fine-tuning.

Metric	Top-1 Acc (%)	Top-5 Acc (%)
Macro Average	67.06	89.27
Micro Average	67.81	89.65

As shown in Table E.1, DGCL achieves 67.81% micro-average Top-1 accuracy and 89.65% micro-average Top-5 accuracy despite the substantial differences in category granularity and data distribution. These results suggest that the learned representations possess promising cross-dataset generalization capability under distribution shifts.

Appendix F. Hyperparameter sensitivity analysis

Impact of LoRA Rank. To analyze the sensitivity of LoRA design choices, we evaluate both the LoRA rank and insertion positions on the Web-Bird dataset. Table F.1 presents the sensitivity analysis of the LoRA rank r . As the rank increases from 4 to 16, the performance gradually improves and reaches the best accuracy of 89.42%. Further increasing the rank to 32 leads to a slight performance degradation. Based on this observation, we adopt $r = 16$ in all experiments.

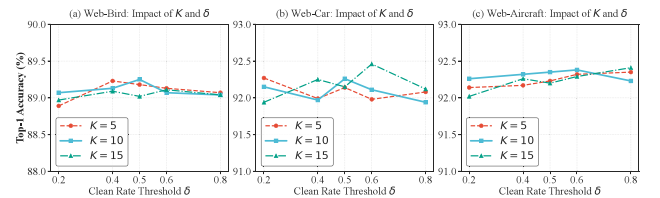


Fig. F.1. Detailed sensitivity analysis of geometric parameters K and δ .

Impact of LoRA Insertion Positions. Furthermore, we evaluate the impact of LoRA insertion positions across transformer layers. Specifically, we compare applying LoRA to only the shallow layers (Layers 1–12), only the deep layers (Layers 13–24), and all layers (Layers 1–24).

Table F.1Sensitivity analysis of the LoRA rank r on the Web-Bird dataset.

LoRA Rank (r)	Trainable Params (M)	Accuracy (%)
$r = 4$	0.79	88.75
$r = 8$	1.57	89.07
$r = 16$	3.15	89.42
$r = 32$	6.29	89.02

Table F.2

Ablation study on LoRA insertion positions on the Web-Bird dataset.

Insertion Position	Trainable Params (M)	Accuracy (%)
Shallow (Layers 1–12)	1.57	89.32
Deep (Layers 13–24)	1.57	89.04
All (Layers 1–24)	3.15	89.42

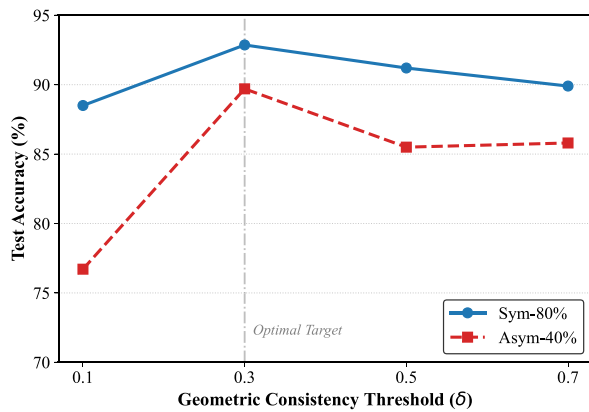


Fig. F.2. Sensitivity analysis of the geometric consistency threshold (δ) on CIFAR-80N under Sym-80% and Asym-40% noise settings (with $K = 10$). A moderately relaxed threshold ($\delta = 0.3$) achieves the best trade-off between noise filtering and hard-positive preservation across both settings.

As shown in Table F.2, applying LoRA to all layers achieves the best accuracy (89.42%). This result suggests that attention recalibration across the entire feature hierarchy is beneficial, as shallow and deep layers provide complementary fine-grained and semantic representations under noisy supervision.

Impact of Geometric Filter Parameters. Fig. F.1 presents the sensitivity analysis of the neighborhood size K and the threshold δ in the geometric filter across three webly supervised datasets. Overall, the performance varies only moderately across different parameter settings. Although the optimal parameter combinations vary slightly across datasets, the unified setting ($K = 10$, $\delta = 0.5$) consistently achieves competitive performance and is therefore adopted in all experiments.

Geometric Consistency Threshold under Extreme Noise. To further evaluate the robustness of the GSC strategy under extreme and open-set noise conditions, we extend the sensitivity analysis of the geometric consistency threshold (δ) to the CIFAR-80N dataset, keeping the neighborhood size fixed at $K = 10$. Fig. F.2 illustrates the test accuracy across varying δ values under both symmetric (Sym-80%) and asymmetric (Asym-40%) noise settings. The results show that $\delta = 0.3$ provides the best overall performance across both noise settings. Compared with the WebFG benchmarks, where a stricter threshold ($\delta = 0.5$) is preferred due to relatively cleaner neighborhood structures, CIFAR-80N contains substantially heavier label corruption and a large proportion of out-of-distribution samples. Under such conditions, enforcing a strict geometric majority criterion may incorrectly reject hard positive samples. Therefore, a moderately relaxed threshold ($\delta = 0.3$) provides a better balance between clean-sample purity and hard-positive recall.

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